Krish Agarwal

ma24btech11014@iith.ac.in | krishagarwal084@gmail.com | +91 9676731537 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.68
FIITJEE	Class XII (TSBIE)	2024	95.30%
Meru International School	Class X (CBSE)	2022	97.83%

SCHOLASTIC ACHIEVEMENTS

Department **Rank 1** in Mathematics and Computing

Awarded **'Excellence In Academics'** for securing highest CGPA in 2nd and 3rd Semester.

Received **A+** grade in Applied Statistics, Ordinary Differential Equations, Number Theory, and Series of Functions

PROJECTS

Substring Matching Using Suffix Trees: | (*under Prof. J Balasubramaniam as a part of DSA Lab*)

- **Implemented Ukkonen's Algorithm** to construct suffix trees in true linear time O(N) relative to text size, utilizing suffix links and skip-and-count optimizations to minimize pointer overhead.
- **Developed an optimized pattern-matching engine** that executes substring queries in O(M + K) time (where M is pattern length and K is the number of occurrences), completely decoupling search performance from database size.
- **Architected a highly efficient text-indexing framework** designed for massive text databases, enabling instantaneous multi-query lookups without the computational overhead of traditional re-scanning methods.

Vectorized Monte Carlo Integration

- Designed and Implemented **interleaved SIMD pseudorandom numbers** using **AVX2** lanes to process **4 independent streams** simultaneously, beating traditional state-dependent PRN generation techniques.
- Implemented support for both parallelized and scalar functions, achieving a **~1.3x speedup for parallelized functions.**
- **Eliminated CPU conversion bottlenecks via IEEE-754 bit-hacking**, directly manipulating double-precision mantissas in-register to bypass costly division instructions and achieve an ultimate compute throughput of **1B+ samples/second**.

Statistical and Data Analysis: (as a part of the Applied Statistics Coursework)

- **Exploratory Data Analysis:** Conducted comprehensive EDA on diverse real-world datasets, leveraging Python and **Pygal** to engineer interactive visualizations that uncovered underlying data distributions and patterns.
- **Statistical Inference & Hypothesis Testing:** Derived and Implemented parametric estimation techniques, including Maximum Likelihood Estimation (**MLE**) and Method of Moments (**MOM**), to model population parameters; designed **t-tests** and **chi-squared tests** to validate statistical hypotheses with strict confidence intervals.
- **Predictive & Correlation Modeling:** Used **linear regression models** to analyze socioeconomic data, utilizing **contingency tables** and correlation matrices to cross-verify feature dependencies and optimize model accuracy.

SKILLS

Programming Languages: C, C++, Python, SQL, MATLAB

Tools and Libraries: NumPy, Pandas, Matplotlib, LaTeX.

RELEVANT COURSES

Applied Mathematics: Numerical Analysis, Probability, Applied Statistics, Ordinary Differential Equations, Convex Optimization, Linear Algebra.

Computer Science: Data Structures and Applications, DSA Lab, Algorithms, Theory of Computation, DBMS-I, Compilers-I, Introduction to Programming.

Pure Mathematics: Calculus, Measure Theory, Number Theory, Transform Techniques.

POSITIONS OF RESPONSIBILITY

Core Member | Cepheid - The Astrophysics Club of IITH

- Used a variety of Python libraries and statistical methods to analyze data in the purview of astrophysics.
- Responsible for conducting sessions on Astronomy.

Teaching Assistant for MA1230 - Series of Functions:

- Assisted in the instruction of an undergraduate real analysis course.
- Prepared assignments on real world applications of series of functions and convergence analysis of series.

Manager | E-CELL IIT Hyderabad

- Helped manage the two flagship events of the club by organizing vendors and funding.

EXTRACURRICULAR

Placed 2nd in Campus-Wide **NPCI Hackathon:**

- Created an viable system that works economically where consumers can use a universal form of reward points, ensuring that smaller stores are not adversely affected by the system.
- Presented a working application prototype where customers can login and select where to spend reward points they've earned.

Core Member at the Writing and Poetry Club at IITH.